

Computer Graphics I

4003-570 / 4005-761

Assignment #3: Clipping

Date posted: March 28th 2013

Due date: April 4th 2013

Submit to the appropriate dropbox on myCourses.

Purpose:

In this assignment, you will implement the Sutherland-Hodgman Clipping Algorithm presented in class.

Goals:

- Improve your understanding of 2D clipping
- Complete your suite of 2D drawing algorithms that can be included as part of your midterm.

2D Programming Environment:

The programming environment that you will use for this assignment, like the previous 2 assignments, is a set of simple object-oriented classes with implementations in both Java and C++. You are free to use either of the implementations. The classes include:

- `simpleCanvas` – a simple 2D canvas that allows the ability to set a pixel.
- `extendedCanvas` – a subclass of `simpleCanvas` that has additional methods used specifically for testing this assignment.
- `clipper` – a simple class that performs the clipping
- `clipTest` – the main function for the application.

You can find the files in the content area of myCourses. Look under the resources category. Note that the distribution contains folders for both C++ and Java versions of the programming environment.

The only class that you will need to modify is the `clipper` class. For this assignment, you will need to complete the method `clipPolygon()`, implementing the Sutherland-Hodgman Clipping Algorithm. The signature for the `clipPolygon` method is:

```
int clipPolygon(int in, float inx[], float iny[], float outx[],  
               float outy[], float x0, float y0, float x1, float y1)
```

```
int clipPolygon(int in, float inx[], float iny[], float outx[],
float outy[], float x0, float y0, float x1, float y1)
```

Where `in` is the number of vertices in the polygon to be clipped, `inx` and `iny` are arrays holding the coordinates of the vertices of the polygon before clipping, `outx` and `outy` are arrays that will hold the coordinates of the vertices of the polygon after clipping, `x0` and `y0` are the coordinates of the lower left corner of the clipping rectangle and `x1` and `y1` are the coordinates of the upper right corner of the clipping rectangle. The function should return the number of vertices in the polygon after clipping. You are free to add additional members and methods to the `clipper` class as you see fit, and you may also include additional auxiliary classes (though this is not required). However, you cannot modify any of the other classes.

Additional Libraries:

The Java implementation of the programming environment makes use of the standard `awt` and `Java2D` classes to access the 2D drawing areas, so no additional libraries or Java files, other than the ones provided by the assignment are required.

However, since C++ does not have a standard 2D graphics environment, you will need some additional libraries if you choose to use the C++ implementation for your submission. The C++ implementation is built on the SFML (Simple and Fast Multimedia Library), a free, cross-platform and lightweight library for performing common multimedia tasks. It can be downloaded at <http://www.sfml-dev.org/download.php>. Version 1.6 of the libraries are used. There are precompiled binaries for Windows, Mac, and Linux. (NOTE: For Windows, there are precompiled binaries for Visual Studio 2005, and Visual Studio 2008. If you are using Visual Studio 2010, a precompiled version of the libraries is available in the content area of myCourses. Look under the resources category.

Please be sure to refer to the installation instructions for your given platform:

- Windows (Visual Studio) - <http://www.sfml-dev.org/tutorials/1.6/start-vc.php>
- Mac OS X -- <http://www.sfml-dev.org/tutorials/1.6/start-osx.php>
- Linux -- <http://www.sfml-dev.org/tutorials/1.6/start-linux.php>

Submission:

Your submission should only include the files for the `clipper` class that you have modified and any other auxiliary classes that you have created. For the Java version, this would mean the file `clipper.java` and for the C++ version, this means the files `clipper.cpp` and `clipper.h`. PLEASE DO NOT SUBMIT code for `simpleCanvas`, `extendedCanvas`, or `clipTest` as you should not be changing the code for these classes.

Grading:

This assignment is worth 10 points distributed as follows:

- 5.1 points for submitting code that compiles, runs, and produces a window.
- 0.7 points for each clipped polygon in the test program. There are 7 such polygons.

Note that the correct output for the test program should look like this:

